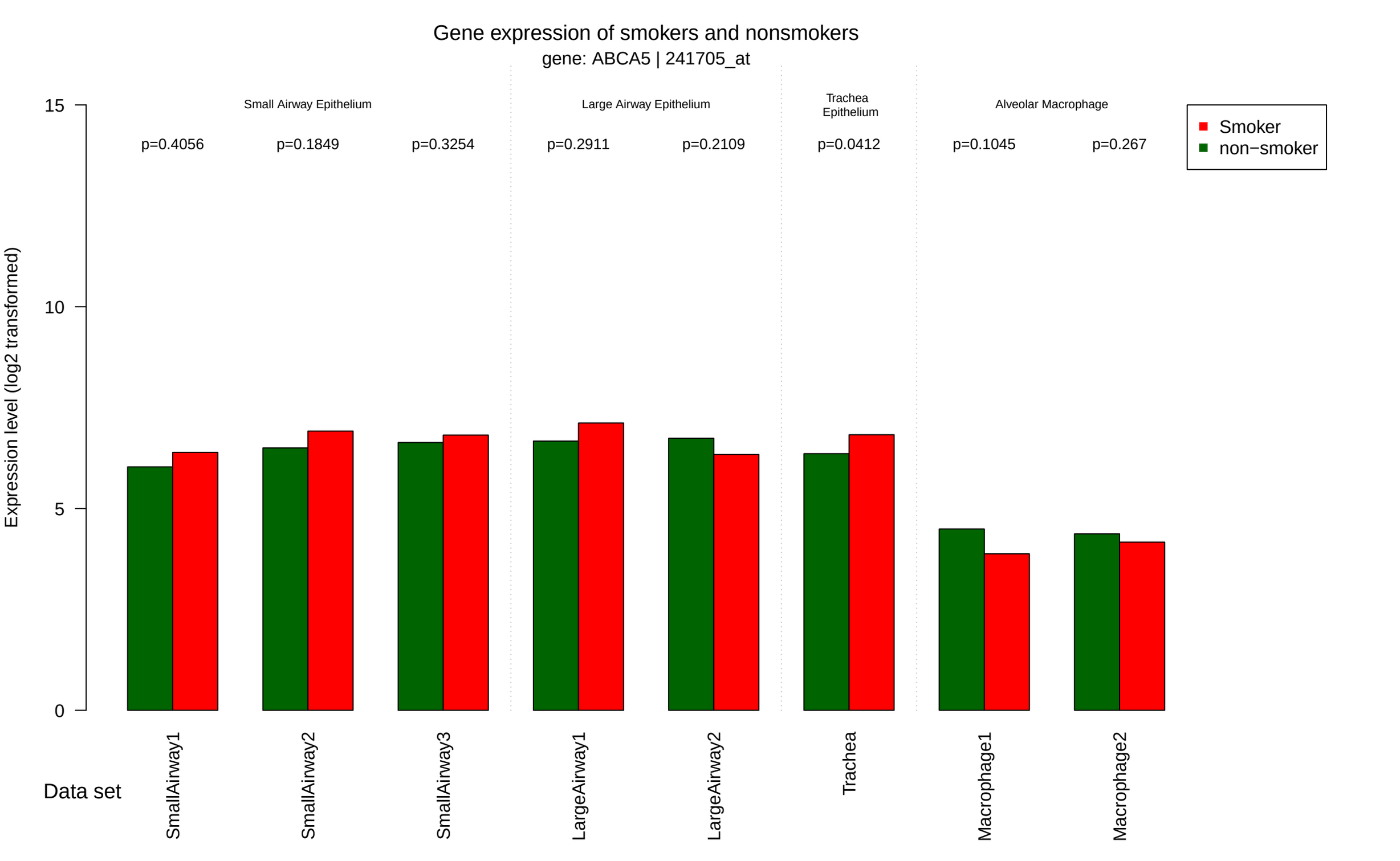
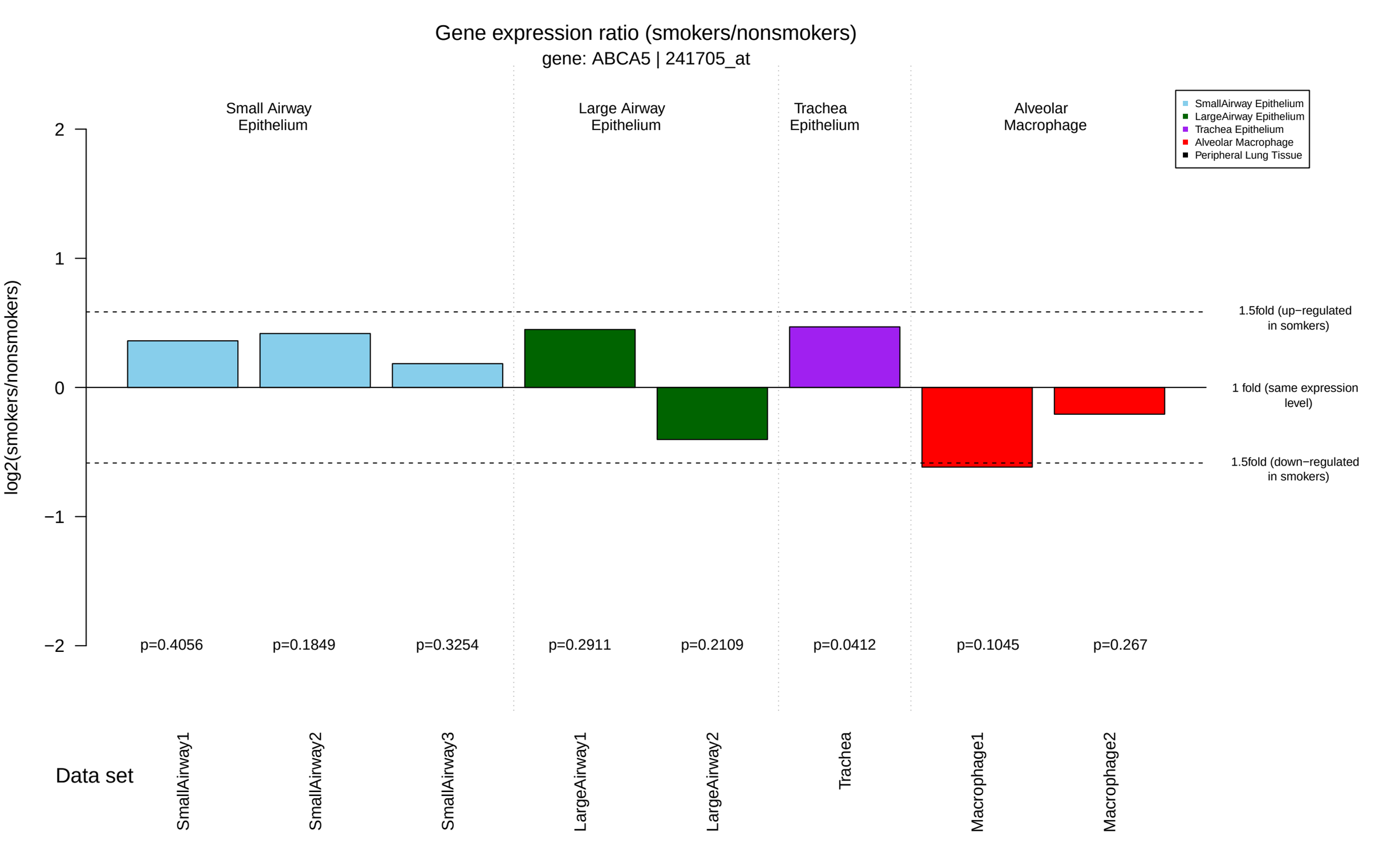
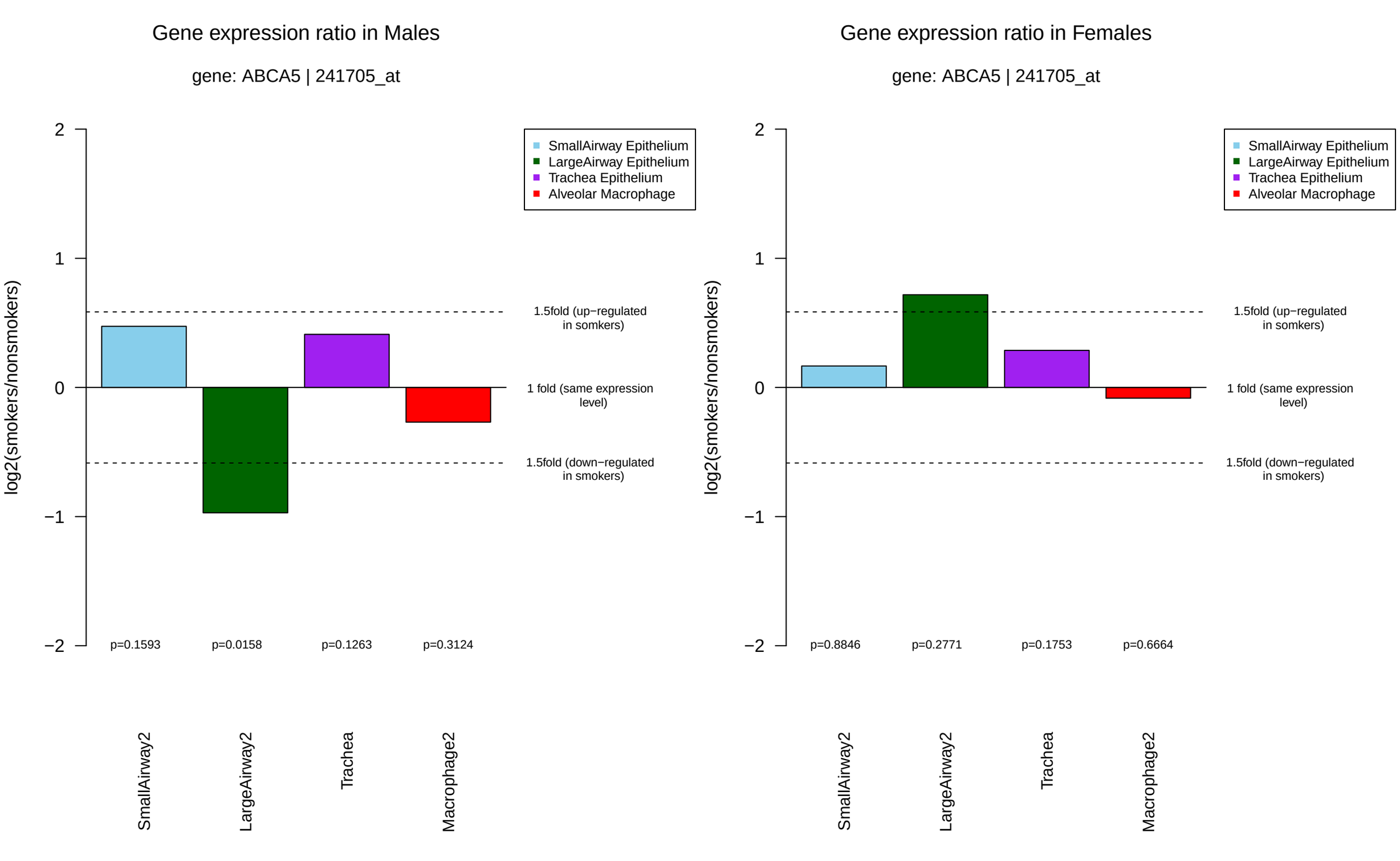
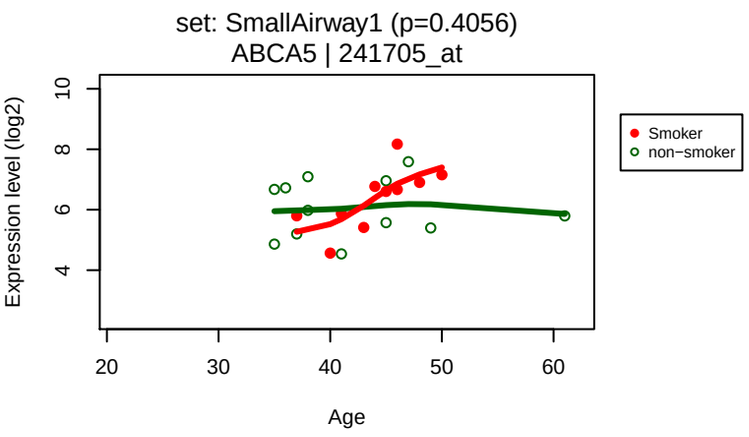
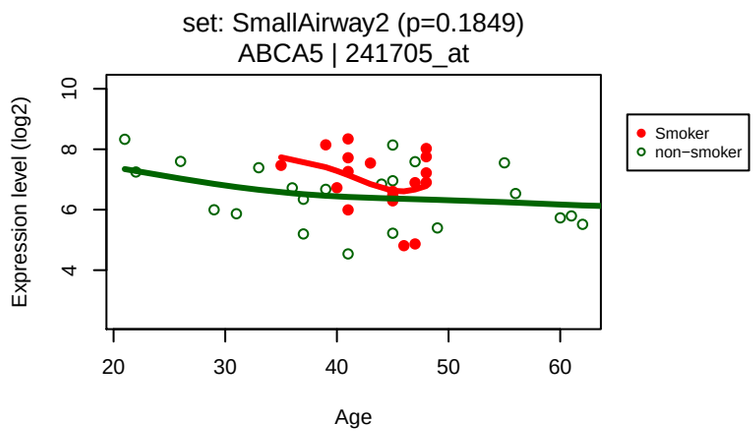
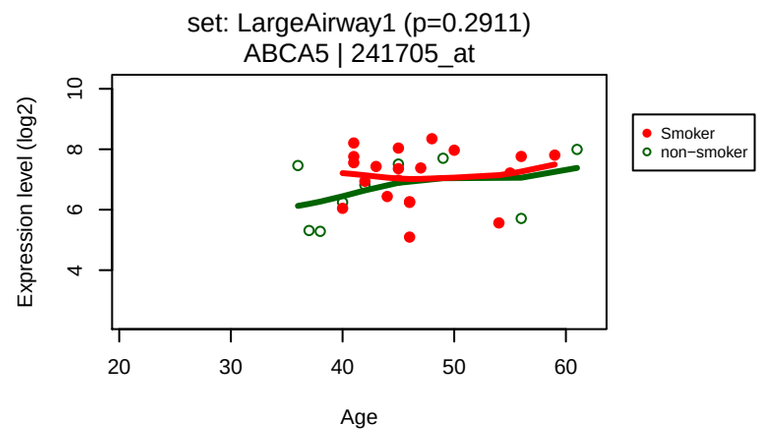

Gene expression of smokers and nonsmokers
gene: ABCA5 | 241705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4056
p=0.1849
p=0.3254
p=0.2911
p=0.2109
p=0.0412
p=0.1045
p=0.267
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

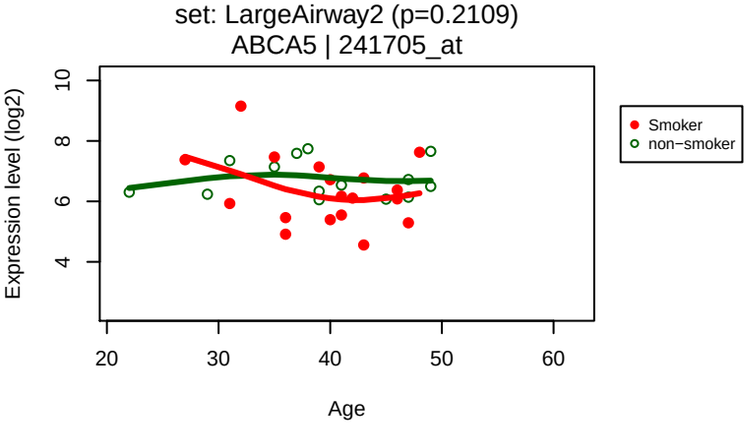

Gene expression ratio (smokers/nonsmokers)
gene: ABCA5 | 241705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4056
p=0.1849
p=0.3254
p=0.2911
p=0.2109
p=0.0412
p=0.1045
p=0.267
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

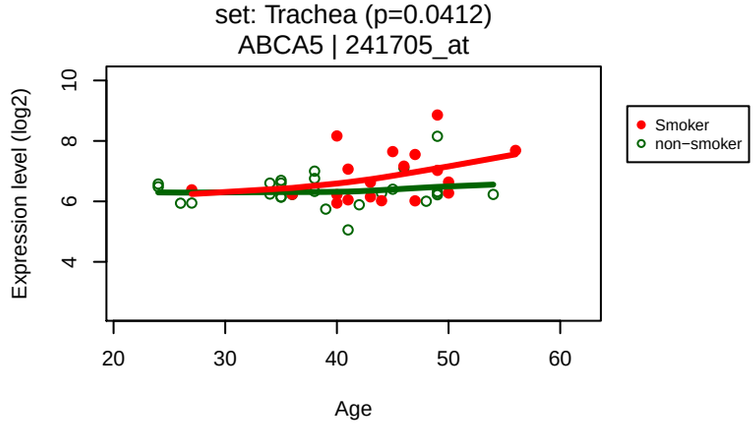

Gene expression ratio in Males
gene: ABCA5 | 241705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1593
p=0.0158
p=0.1263
p=0.3124
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCA5 | 241705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8846
p=0.2771
p=0.1753
p=0.6664
SmallAirway2
LargeAirway2
Trachea
Macrophage2

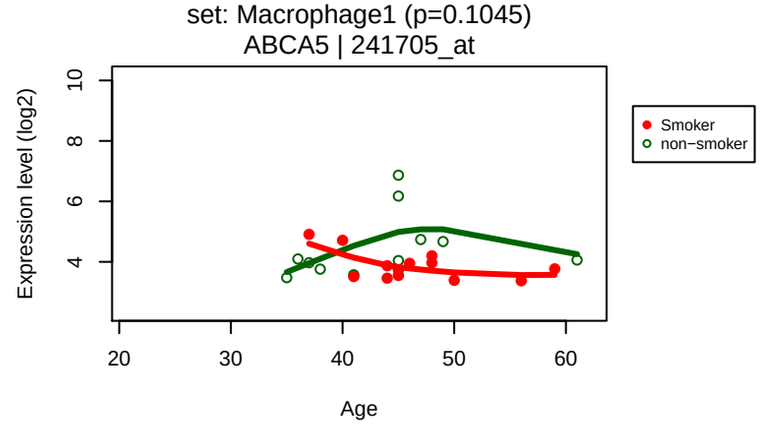

set: SmallAirway1 (p=0.4056)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker

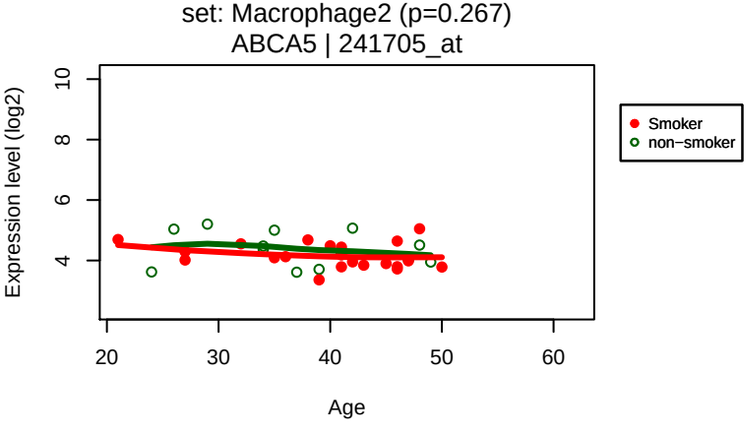

set: SmallAirway2 (p=0.1849)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker

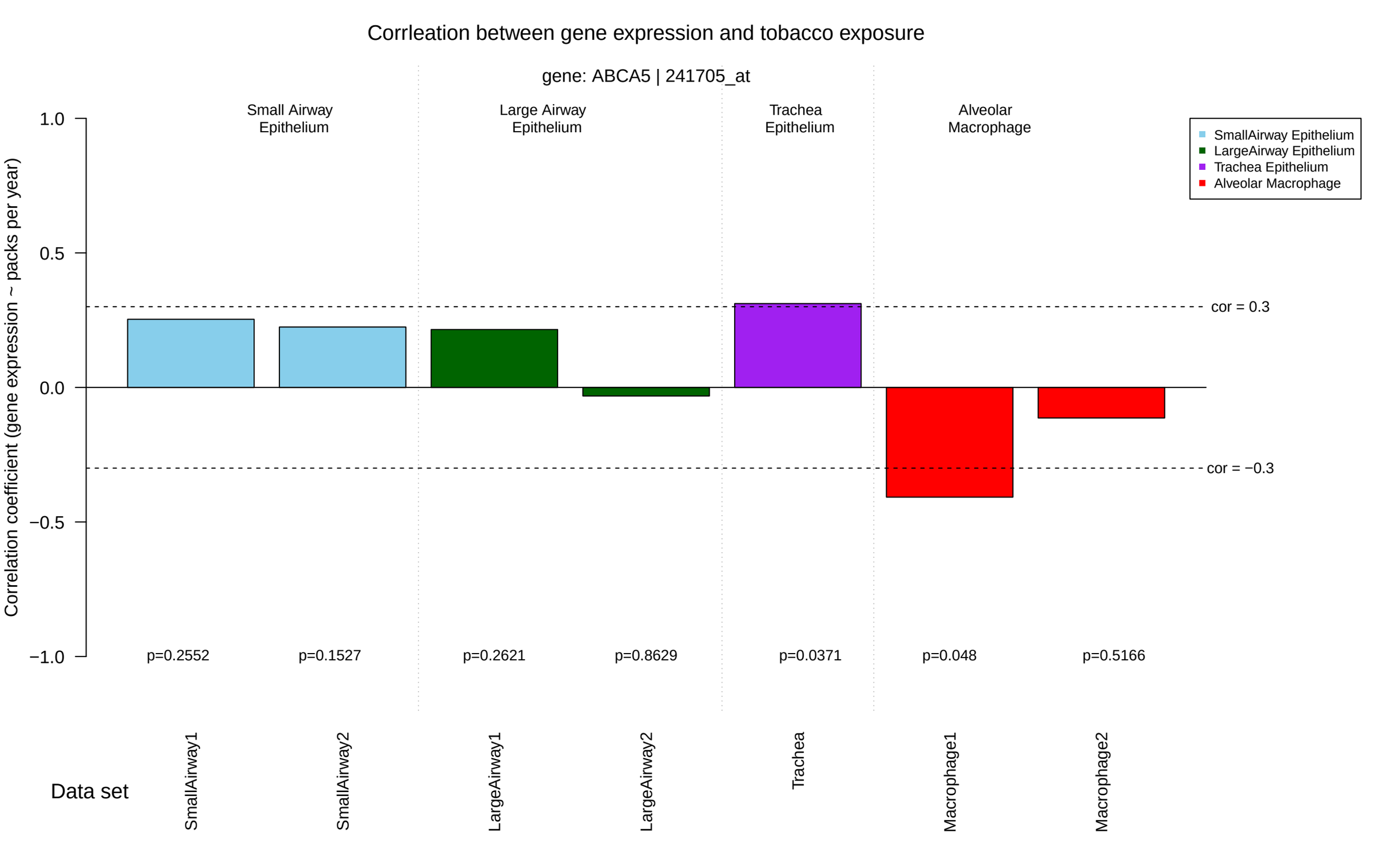

set: LargeAirway1 (p=0.2911)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2109)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0412)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1045)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.267)
ABCA5 | 241705_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCA5 | 241705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2552
p=0.1527
p=0.2621
p=0.8629
p=0.0371
p=0.048
p=0.5166
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2